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Lucey

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(54) **SYSTEM AND METHOD FOR INDIVIDUAL
PLAYER AND TEAM SIMULATION**

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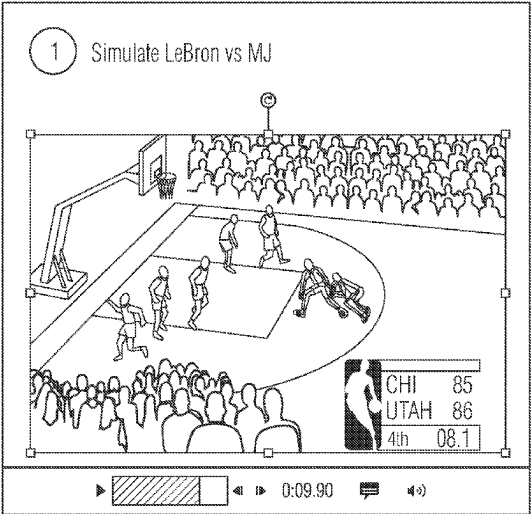
See application file for complete search history.

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ABSTRACT

A computing system retrieves historical event data for a plurality of games in a league. The historical event data includes (x,y) coordinates of players within each game and game context data. The computing system learns one or more attributes of each team in each game and each player on each team in each game. The computing system receives a request to simulate a play in a historical game. The request includes substituting a player that was in the play with a target player that was not in the play. The computing system simulates the play with the target player in place of the player based on the one or more attributes learned by the computing system. The computing system generates a graphical representation of the simulation.

17 Claims, 5 Drawing Sheets



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* cited by examiner

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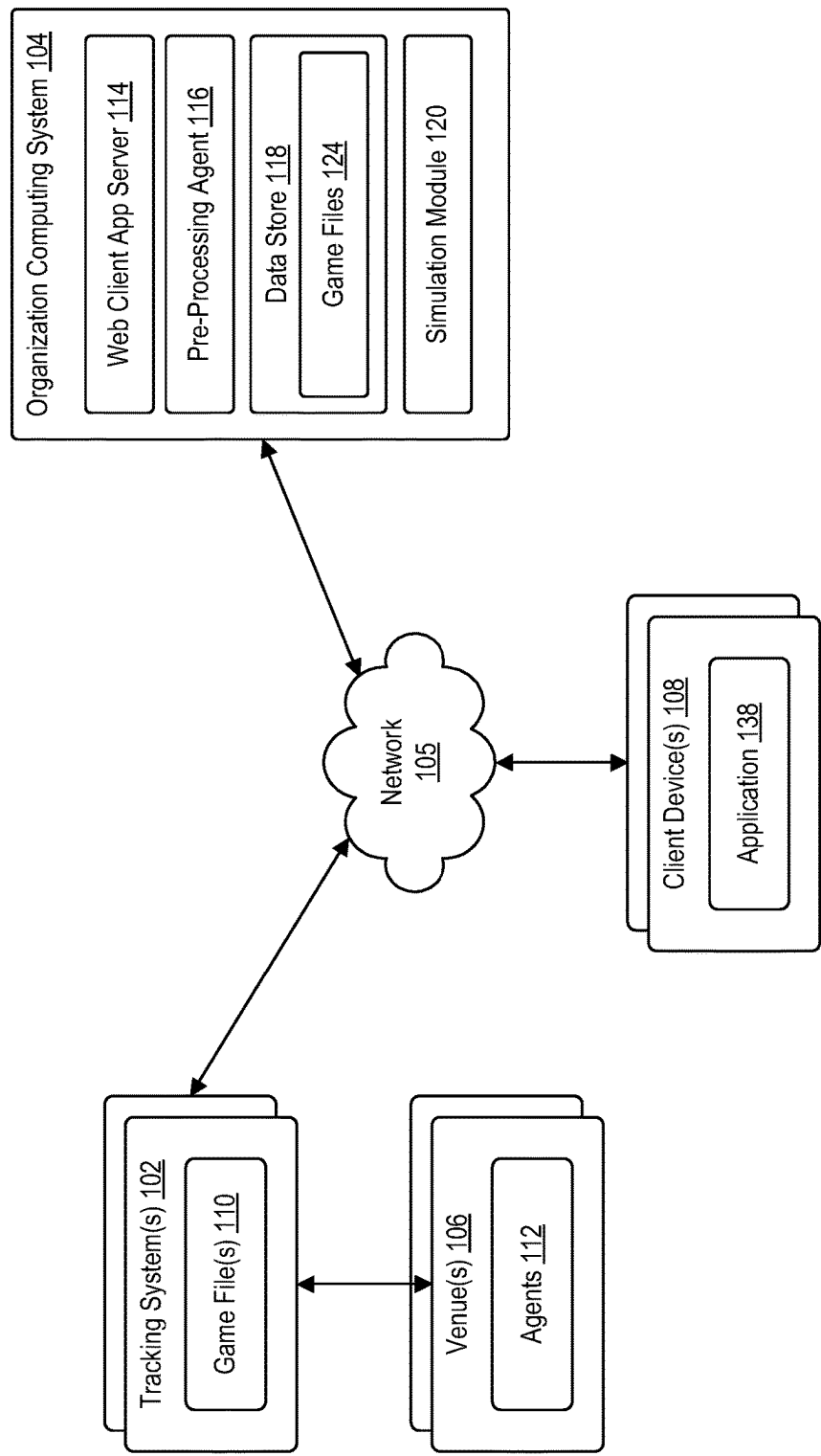


FIG. 1

200

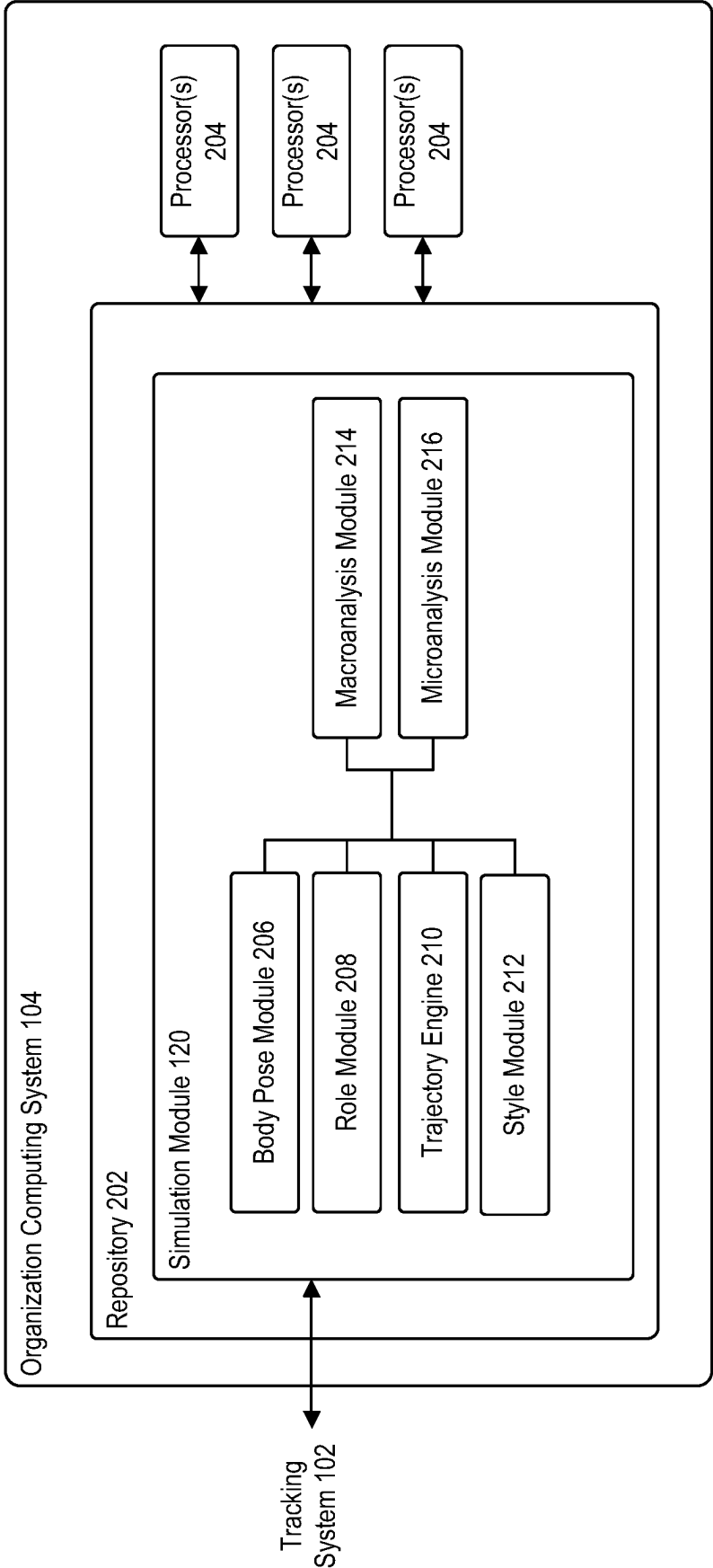


FIG. 2

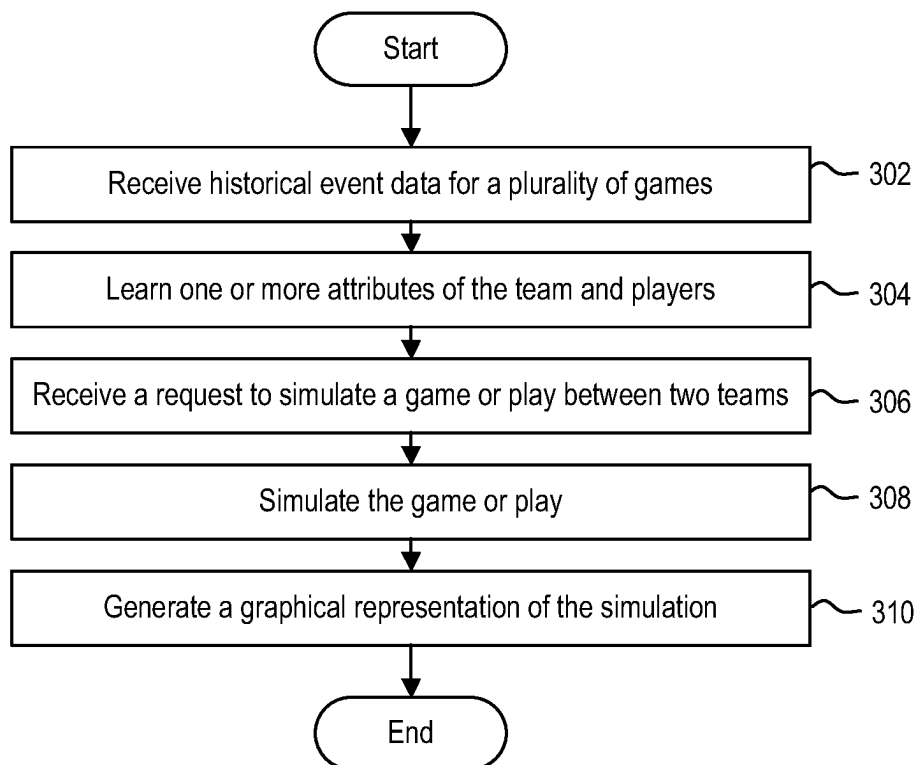
300

FIG. 3

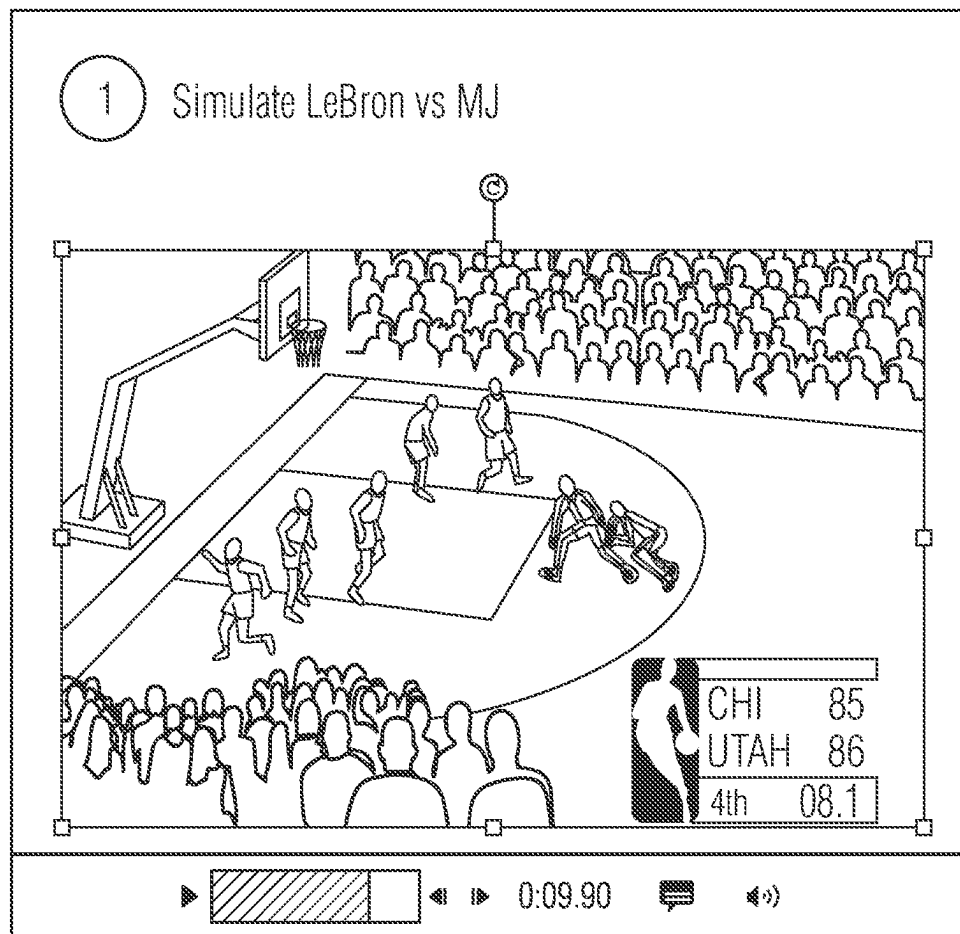
400

FIG. 4

FIG. 5A

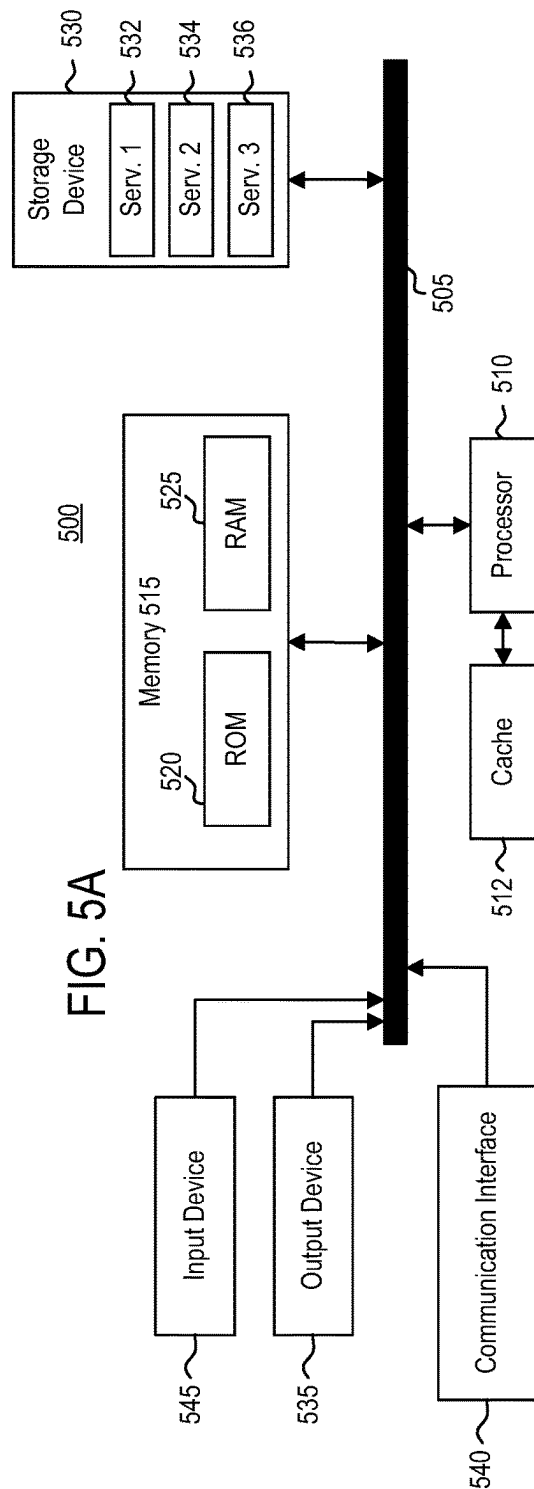
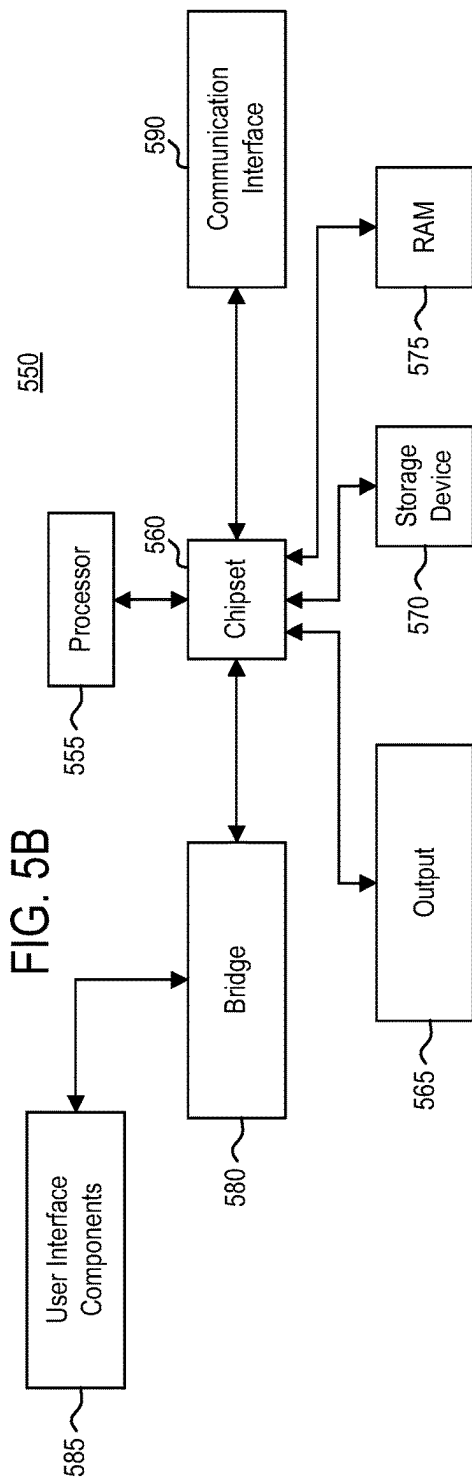


FIG. 5B



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SYSTEM AND METHOD FOR INDIVIDUAL PLAYER AND TEAM SIMULATION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of, and claims the benefit of priority to, U.S. application Ser. No. 17/660,980, filed Apr. 27, 2022, which claims priority to U.S. Provisional Application Ser. No. 63/180,168, filed Apr. 27, 2021, the entireties of each of which are incorporated herein by reference.

FIELD OF THE DISCLOSURE

The present disclosure generally relates to system and method for individual player and team simulation using historical tracking data.

BACKGROUND

In professional sports, fans and commentators alike incessantly argue about how players compare players from different generations. One of the more famous debates centers around that of LeBron James and Michael Jordan. Both players have dominated their respective generations and are among the two most mentioned players considered to be the “best of all time.” While fans and commentators can debate the issue, there is currently no mechanism to quantify or visualize a matchup between the two players.

SUMMARY

In some embodiments, a method is disclosed herein. A computing system retrieves historical event data for a plurality of games in a league. The historical event data includes (x,y) coordinates of players within each game and game context data. The computing system learns one or more attributes of each team in each game and each player on each team in each game. The computing system receives a request to simulate a play in a historical game. The request includes substituting a player that was in the play with a target player that was not in the play. The computing system simulates the play with the target player in place of the player based on the one or more attributes learned by the computing system. The computing system generates a graphical representation of the simulation.

In some embodiments, a non-transitory computer readable medium is disclosed herein. The non-transitory computer readable medium includes one or more sequences of instructions, which, when executed by a processor, causes a computing system to perform operations. The operations include retrieving, by the computing system, historical event data for a plurality of games in a league. The historical event data include player information within each game and game context data. The operations further include learning, by the computing system, one or more attributes of each team in each game and each player on each team in each game. The operations further include receiving, by the computing system, a request to simulate a play in a historical game. The request includes substituting a player that was in the play with a target player that was not in the play. The operations further include simulating, by the computing system, the play with the target player in place of the player based on the one or more attributes learned by the computing system. The operations further include generating, by the computing system, a graphical representation of the simulation.

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In some embodiments, a system is disclosed herein. The system includes a processor and a memory. The memory has programming instructions stored thereon, which, when executed by the processor, causes the system to perform operations. The operations include retrieving historical event data for a plurality of games in a league. The historical event data include player information within each game and game context data. The operations further include learning, by a simulation module, one or more attributes of each team in each game and each player on each team in each game. The operations further include receiving a request to simulate a play in a historical game. The request includes substituting a player that was in the play with a target player that was not in the play. The operations further include simulating, by the simulation module, the play with the target player in place of the player based on the one or more attributes learned by the simulation module. The operations further include generating a graphical representation of the simulation.

BRIEF DESCRIPTION OF THE DRAWINGS

So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

FIG. 1 is a block diagram illustrating a computing environment, according to example embodiments.

FIG. 2 is a block diagram illustrating the organization computing system of FIG. 1, according to example embodiments.

FIG. 3 is a flow diagram illustrating a method of simulating a play, according to example embodiments.

FIG. 4 illustrates an example graphical user interface according to example embodiments.

FIG. 5A is a block diagram illustrating a computing device, according to example embodiments.

FIG. 5B is a block diagram illustrating a computing device, according to example embodiments.

To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially utilized on other embodiments without specific recitation.

DETAILED DESCRIPTION

One or more techniques provided herein provides a system and method for replaying a play in a given event by changing the players involved in the play and simulating what the alternative outcome would be. For example, one of the more infamous plays in sports’ history involves Michael Jordan hitting the game-winning jump shot against the Utah Jazz over Byron Russell in game 6 of the 1998 NBA Finals. Using the one or more techniques provided herein, the present system is able to replace Byron Russell with another player, such as, but not limited to Larry Bird, Wilt Chamberlain, Magic Johnson, LeBron James, or Kawhi Leonard.

To achieve such functionality, the present system may utilize historical tracking data. Such historical tracking data may include the (x, y) coordinates of each player during the

course of the game and/or body-pose information of each player during the course of the game. Further, to account for differences in errors, the present system may utilize one or more model adaptation techniques (e.g., transfer learning) to ensure that differences between eras may be normalized to increase the accuracy of the simulation.

While the present discussion is provided in the context of both soccer and basketball, those skilled in the art readily understand that such functionality may be extended to other sports.

FIG. 1 is a block diagram illustrating a computing environment 100, according to example embodiments. Computing environment 100 may include tracking system 102, organization computing system 104, and one or more client devices 108 communicating via network 105.

Network 105 may be of any suitable type, including individual connections via the Internet, such as cellular or Wi-Fi networks. In some embodiments, network 105 may connect terminals, services, and mobile devices using direct connections, such as radio frequency identification (RFID), near-field communication (NFC), Bluetooth™, low-energy Bluetooth™ (BLE), Wi-Fi™, ZigBee™, ambient backscatter communication (ABC) protocols, USB, WAN, or LAN. Because the information transmitted may be personal or confidential, security concerns may dictate one or more of these types of connection be encrypted or otherwise secured. In some embodiments, however, the information being transmitted may be less personal, and therefore, the network connections may be selected for convenience over security.

Network 105 may include any type of computer networking arrangement used to exchange data or information. For example, network 105 may be the Internet, a private data network, virtual private network using a public network and/or other suitable connection(s) that enables components in computing environment 100 to send and receive information between the components of environment 100.

Tracking system 102 may be positioned in a venue 106. For example, venue 106 may be configured to host a sporting event that includes one or more agents 112. Tracking system 102 may be configured to record the motions of all agents (i.e., players) on the playing surface, as well as one or more other objects of relevance (e.g., ball, referees, etc.). In some embodiments, tracking system 102 may be an optically-based system using, for example, a plurality of fixed cameras. For example, a system of six stationary, calibrated cameras, which project the three-dimensional locations of players and the ball onto a two-dimensional overhead view of the court may be used. In some embodiments, tracking system 102 may be a radio-based system using, for example, radio frequency identification (RFID) tags worn by players or embedded in objects to be tracked. Generally, tracking system 102 may be configured to sample and record, at a high frame rate (e.g., 25 Hz). Tracking system 102 may be configured to store at least player identity and positional information (e.g., (x, y) position) for all agents and objects on the playing surface for each frame in a game file 110.

Game file 110 may be augmented with other event information corresponding to event data, such as, but not limited to, game event information (pass, made shot, turnover, etc.) and context information (current score, time remaining, etc.).

Tracking system 102 may be configured to communicate with organization computing system 104 via network 105. Organization computing system 104 may be configured to manage and analyze the data captured by tracking system 102. Organization computing system 104 may include at

least a web client application server 114, a pre-processing agent 116, a data store 118, and a simulation module 120. Each of pre-processing agent 116 and simulation module 120 may be comprised of one or more software modules. The one or more software modules may be collections of code or instructions stored on a media (e.g., memory of organization computing system 104) that represent a series of machine instructions (e.g., program code) that implements one or more algorithmic steps. Such machine instructions may be the actual computer code the processor of organization computing system 104 interprets to implement the instructions or, alternatively, may be a higher level of coding of the instructions that is interpreted to obtain the actual computer code. The one or more software modules may also include one or more hardware components. One or more aspects of an example algorithm may be performed by the hardware components (e.g., circuitry) itself, rather as a result of the instructions.

Data store 118 may be configured to store one or more game files 124. Each game file 124 may include spatial event data and non-spatial event data. For example, spatial event data may correspond to raw data captured from a particular game or event by tracking system 102. Non-spatial event data may correspond to one or more variables describing the events occurring in a particular match without associated spatial information. For example, non-spatial event data may correspond to each play-by-play event in a particular match. In some embodiments, non-spatial event data may be derived from spatial event data. For example, pre-processing agent 116 may be configured to parse the spatial event data to derive play-by-play information. In some embodiments, non-spatial event data may be derived independently from spatial event data. For example, an administrator or entity associated with organization computing system may analyze each match to generate such non-spatial event data. As such, for purposes of this application, event data may correspond to spatial event data and non-spatial event data.

In some embodiments, each game file 124 may further include the home and away team box scores. For example, the home and away teams' box scores may include the number of team assists, fouls, rebounds (e.g., offensive, defensive, total), steals, and turnovers at each time, t, during gameplay. In some embodiments, each game file 124 may further include a player box score. For example, the player box score may include the number of player assists, fouls, rebounds, shot attempts, points, free-throw attempts, free-throws made, blocks, turnovers, minutes played, plus/minus metric, game started, and the like. Although the above metrics are discussed with respect to basketball, those skilled in the art readily understand that the specific metrics may change based on sport. For example, in soccer, the home and away teams' box scores may include shot attempts, assists, crosses, shots, and the like.

Pre-processing agent 116 may be configured to process data retrieved from data store 118. For example, pre-processing agent 116 may be configured to generate one or more sets of information that may be used to train simulation module 120.

Simulation module 120 may be configured to simulate play between various players or teams based on historical tracking data of the players and/or teams. To do so, simulation module 120 may analyze historical games associated with a given league. For example, simulation module 120 may utilize data generated by AutoSTATS to analyze historical game information.

Once simulation module 120 generates or learns various data associated with the historical game information (e.g.,

body pose information, role information, trajectory information, and/or style information), simulation module **120** may be able to simulate behavior a game or a play between teams or players.

In some embodiments, simulation module **120** may be able to simulate a game or play between teams or players even across generations. For example, simulation module **120** may be able to simulate a play between LeBron James and Michael Jordan, despite the fact that the two players have never played an NBA game against each other. To accurately simulate the performance, simulation module **120** may utilize a transfer learning process. In this manner, simulation module **120** may be able to transfer knowledge of a player in a first era to the course of play in a second era. In other words, simulation module **120** may account for differences of play between eras. In this manner, simulation module **120** can adapt players of different eras to predict how they would play in another era.

Client device **108** may be in communication with organization computing system **104** via network **105**. Client device **108** may be operated by a user. For example, client device **108** may be a mobile device, a tablet, a desktop computer, or any computing system having the capabilities described herein. Users may include, but are not limited to, individuals such as, for example, subscribers, clients, prospective clients, or customers of an entity associated with organization computing system **104**, such as individuals who have obtained, will obtain, or may obtain a product, service, or consultation from an entity associated with organization computing system **104**.

Client device **108** may include at least application **138**. Application **138** may be representative of a web browser that allows access to a website or a stand-alone application. Client device **108** may access application **138** to access one or more functionalities of organization computing system **104**. Client device **108** may communicate over network **105** to request a webpage, for example, from web client application server **114** of organization computing system **104**. For example, client device **108** may be configured to execute application **138** to request a simulation from simulation module **120** and/or view a simulation generated by simulation module **120**.

FIG. 2 is a block diagram **200** illustrating organization computing system **104**, according to example embodiments. As shown, organization computing system **104** includes repository **202** and one or more computer processors **204**.

Repository **202** may be representative of any type of storage unit and/or device (e.g., a file system, database, collection of tables, or any other storage mechanism) for storing data. Further, repository **202** may include multiple different storage units and/or devices. The multiple different storage units and/or devices may or may not be of the same type or located at the same physical site. As shown, repository **202** includes at least simulation module **120**.

Simulation module **120** may include at least body pose module **206**, role module **208**, trajectory engine **210**, style module **212**, macroanalysis module **214**, and microanalysis module **216**. Each of body pose module **206**, role module **208**, trajectory engine **210**, style module **212**, macroanalysis module **214**, and microanalysis module **216** may be comprised of one or more software modules. The one or more software modules are collections of code or instructions stored on a media (e.g., memory of organization computing system **104**) that represent a series of machine instructions (e.g., program code) that implements one or more algorithmic steps. Such machine instructions may be the actual computer code the processor of organization computing

system **104** interprets to implement the instructions or, alternatively, may be a higher level of coding of the instructions that are interpreted to obtain the actual computer code. The one or more software modules may also include one or more hardware components. One or more aspects of an example algorithm may be performed by the hardware components (e.g., circuitry) itself, rather than as a result of the instructions.

As shown, simulation module **120** may be configured to receive tracking data and/or event data from tracking system **102**. In some embodiments, the tracking data may be representative of raw (x, y) position information of each player on the playing surface during a course of the game. In some embodiments, event data may indicate various attributes of the game at various points, such as, but not limited to, at one or more points during the game: the players in the game, the players out of the game, the current score, the current time, player statistics at the current time, and the like. In some embodiments, the tracking data may be derived from broadcast video data received from tracking system **102**.

Body pose module **206** may be configured to generate one or more metrics related to the body pose of at least one or more of a player throughout a game based on the tracking data and/or event data. In some embodiments, body pose module **206** may generate body pose information based on event data captured by tracking system **102**. In some embodiments, body pose module **206** may generate body pose information from a broadcast stream provided by a broadcast provider. Body pose module **206** may be able to identify, for example, shooter start position and angle, run type (e.g., stutter and speed), shot initiation (e.g., body lean angle, upper body angle, hip orientation, kicking arm position, shoulder alignment, etc.), and the like. Additionally, the raw positions of the body-positions in 2D or 3D which appear as a skeleton can be used to detect and correlate specific key actions in sports. Generally, body pose module **206** may be representative of a body pose module disclosed in U.S. application Ser. No. 16/804,964, which is hereby incorporated by reference in its entirety.

Role module **208** may be configured to detect a role in each player of each historical game based on the tracking data and/or event data. For example, role module **208** may be configured to predict various different aspects of team and player functions within a team. For example, role module **208** may utilize an ensemble of models that may work in conjunction to learn the role associated with a given player. In some embodiments, role module **208** may utilize event data to make such determination. Generally, role module **208** may be representative of role prediction platform disclosed in U.S. application Ser. No. 17/167,400, which is hereby incorporated by reference in its entirety.

Trajectory engine **210** may be configured to learn trajectories of each player in each historical game based on the tracking data and/or event data. For example, trajectory engine **210** may be configured to predict the trajectory of one or more agents given one or more historical trajectory points. For example, given an agent's coordinates up to a time t_q , trajectory engine **210** may use at least the agent's coordinates up to time t_q predict the agent's coordinates up to time t_f where t_f is after t_q , where q represents some end-time between (e.g., $(1 \dots n)$) and f represents some future time, after q (e.g., $(n+1)$). Generally, trajectory engine **210** may be representative of trajectory agent disclosed in U.S. application Ser. No. 16/254,037, which is hereby incorporated by reference in its entirety.

Style module **212** may be configured to learn the various styles and content of each team in each historical game

based on the tracking data and/or event data. Content of a given play may be referred to as the “what” of the play, independent of the exact specifics of how a team executes the play. In contrast, style may be referred to as the “how” of the play, which captures the various ways a given play can evolve. Generally, style module **212** may be representative of team prediction agent disclosed in U.S. application Ser. No. 16/870,170, which is hereby incorporated by reference in its entirety.

Macroanalysis module **214** may be configured to perform one or more macro analyses based on one or more of the learned body pose information, the learned role information, the learned trajectory information, or the learned style information. For example, macroanalysis module **214** may be configured to utilize one or more of the learned body pose information, the learned role information, the learned trajectory information, or the learned style information to answer counterfactual questions, such as, determining where Michael Jordan would be drafted in an upcoming NBA draft based on Michael Jordan’s collegiate statistics at the University of North Carolina. In such example, macroanalysis module **214** may leverage the prediction model disclosed in U.S. application Ser. No. 17/449,694, which is hereby incorporated by reference in its entirety.

In another example, macroanalysis module **214** may be configured to utilize one or more of the learned body pose information, the learned role information, the learned trajectory information, or the learned style information to answer counterfactual questions, such as, determining the performance of Pele in the English Premier League. In such example, macroanalysis module **214** may leverage the prediction models disclosed in U.S. Provisional Application No. 63/201,898 and U.S. Provisional Application No. 63/267,062, which are both hereby incorporated by reference in their entireties.

Microanalysis module **216** may be configured to perform one or more micro analyses based on one or more of the learned body pose information, the learned role information, the learned trajectory information, or the learned style information. For example, microanalysis module **216** may be configured to simulate a play or a game by replacing a player in the play or game with another player based on one or more of the learned body pose information, the learned role information, the learned trajectory information, or the learned style information. For example, microanalysis module **216** may leverage one or more graph neural networks disclosed in U.S. application Ser. No. 17/649,970, which is hereby incorporated by reference in its entirety, to model defensive behavior of the player.

FIG. **3** is a flow diagram illustrating a method **300** of simulating a game or play, according to example embodiments. Method **300** may begin at step **302**.

At step **302**, organization computing system **104** may retrieve historical event data for a plurality of games in a league. In some embodiments, the historical event data may include the (x, y) coordinates of all players at all points in the game. In some embodiments, the historical event data may include game context data, such as the time of each movement, score at each movement, and the like. Generally, the plurality of games may span several eras. Taking basketball for example, the plurality of games may span the 1960s to current day.

At step **304**, organization computing system **104** may learn various attributes of the teams and players in each game. For example, simulation module **120** may learn various attributes of the teams and players in each game, such that simulation module **120** can accurately simulate

performance between players and/or teams that may not have ever played each other or did not play each other given the current game context.

At step **306**, organization computing system **104** may receive a request to simulate a game or play between two teams. In some embodiments, simulation module **120** may receive the request from client device **108** via application **138** executing thereon. In some embodiments, the request may include replacing one player in a game or player with a different player that is otherwise not involved in the current play of the game. For example, continuing with the above, Lebron James may replace Michael Jordan in the game winning play of Game 6 of the 1998 NBA Finals.

At step **308**, organization computing system **104** may simulate the game or play. For example, simulation module **120** may simulate the play between Lebron James and Byron Russell based on learned body pose information, role information, trajectory information, playing style and playing style information. In some embodiments, simulation module **120** may further take into account the difference between eras. For example, simulation module **120** may adapt Lebron James’ playing style to the NBA playing style of 1998.

At step **310**, organization computing system **104** may generate a graphical representation of the simulation. For example, simulation module **120** may generate a visual corresponding to Lebron James defending Michael Jordan in the game winning play of the 1998 NBA Finals. In some embodiments, simulation module **120** may provide a visual indication of each player’s body pose information.

FIG. **4** illustrates an example graphical user interface (GUI) **400** according to example embodiments. As shown, graphical user interface **400** may correspond to a simulation of Lebron James guarding Michael Jordan in the game winning play of the 1998 NBA Finals. For example, as shown, rather than Byron Russel guarding Michael Jordan in the game winning player, a user has requested that Lebron James be guarding Byron Russel. As shown, GUI **400** may include a skeletal representation of Lebron James guarding a skeletal representation of Michael Jordan. Based on learned information about Lebron James (e.g., one or more of the learned body pose information, the learned role information, the learned trajectory information, or the learned style information), simulation module **120** may simulate the game winning play as if Lebron James was guarding Michael Jordan in that situation rather than Byron Russel.

Although GUI **400** illustrates a scenario in which the defensive player is replaced with another player, those skilled in the art understand that, instead of replacing Bryon Russel, a user may instead replace Michael Jordan with Lebron James. In this manner, based on learned information about Lebron James (e.g., one or more of the learned body pose information, the learned role information, the learned trajectory information, or the learned style information), simulation module **120** may simulate the game winning play as if Lebron James had possession against Byron Russel instead of Michael Jordan.

FIG. **5A** illustrates a system bus architecture of computing system **500**, according to example embodiments. System **500** may be representative of at least a portion of organization computing system **104**. One or more components of system **500** may be in electrical communication with each other using a bus **505**. System **500** may include a processing unit (CPU or processor) **510** and a system bus **505** that couples various system components including the system memory **515**, such as read only memory (ROM) **520** and random access memory (RAM) **525**, to processor **510**.

System **500** may include a cache of high-speed memory connected directly with, in close proximity to, or integrated as part of processor **510**. System **500** may copy data from memory **515** and/or storage device **530** to cache **512** for quick access by processor **510**. In this way, cache **512** may provide a performance boost that avoids processor **510** delays while waiting for data. These and other modules may control or be configured to control processor **510** to perform various actions. Other system memory **515** may be available for use as well. Memory **515** may include multiple different types of memory with different performance characteristics. Processor **510** may include any general purpose processor and a hardware module or software module, such as service 1 **532**, service 2 **534**, and service 3 **536** stored in storage device **530**, configured to control processor **510** as well as a special-purpose processor where software instructions are incorporated into the actual processor design. Processor **510** may essentially be a completely self-contained computing system, containing multiple cores or processors, a bus, memory controller, cache, etc. A multi-core processor may be symmetric or asymmetric.

To enable user interaction with the computing system **500**, an input device **545** may represent any number of input mechanisms, such as a microphone for speech, a touch-sensitive screen for gesture or graphical input, keyboard, mouse, motion input, speech and so forth. An output device **535** may also be one or more of a number of output mechanisms known to those of skill in the art. In some instances, multimodal systems may enable a user to provide multiple types of input to communicate with computing system **500**. Communications interface **540** may generally govern and manage the user input and system output. There is no restriction on operating on any particular hardware arrangement and therefore the basic features here may easily be substituted for improved hardware or firmware arrangements as they are developed.

Storage device **530** may be a non-volatile memory and may be a hard disk or other types of computer readable media which may store data that are accessible by a computer, such as magnetic cassettes, flash memory cards, solid state memory devices, digital versatile disks, cartridges, random access memories (RAMs) **525**, read only memory (ROM) **520**, and hybrids thereof.

Storage device **530** may include services **532**, **534**, and **536** for controlling the processor **510**. Other hardware or software modules are contemplated. Storage device **530** may be connected to system bus **505**. In one aspect, a hardware module that performs a particular function may include the software component stored in a computer-readable medium in connection with the necessary hardware components, such as processor **510**, bus **505**, output device **535** (e.g., display), and so forth, to carry out the function.

FIG. 5B illustrates a computer system **550** having a chipset architecture that may represent at least a portion of organization computing system **104**. Computer system **550** may be an example of computer hardware, software, and firmware that may be used to implement the disclosed technology. System **550** may include a processor **555**, representative of any number of physically and/or logically distinct resources capable of executing software, firmware, and hardware configured to perform identified computations. Processor **555** may communicate with a chipset **560** that may control input to and output from processor **555**. In this example, chipset **560** outputs information to output **565**, such as a display, and may read and write information to storage device **570**, which may include magnetic media, and solid state media, for example. Chipset **560** may also read

data from and write data to storage device **575** (e.g., RAM). A bridge **580** for interfacing with a variety of user interface components **585** may be provided for interfacing with chipset **560**. Such user interface components **585** may include a keyboard, a microphone, touch detection and processing circuitry, a pointing device, such as a mouse, and so on. In general, inputs to system **550** may come from any of a variety of sources, machine generated and/or human generated.

Chipset **560** may also interface with one or more communication interfaces **590** that may have different physical interfaces. Such communication interfaces may include interfaces for wired and wireless local area networks, for broadband wireless networks, as well as personal area networks. Some applications of the methods for generating, displaying, and using the GUI disclosed herein may include receiving ordered datasets over the physical interface or be generated by the machine itself by processor **555** analyzing data stored in storage device **570** or storage device **575**. Further, the machine may receive inputs from a user through user interface components **585** and execute appropriate functions, such as browsing functions by interpreting these inputs using processor **555**.

It may be appreciated that example systems **500** and **550** may have more than one processor **510** or be part of a group or cluster of computing devices networked together to provide greater processing capability.

While the foregoing is directed to embodiments described herein, other and further embodiments may be devised without departing from the basic scope thereof. For example, aspects of the present disclosure may be implemented in hardware or software or a combination of hardware and software. One embodiment described herein may be implemented as a program product for use with a computer system. The program(s) of the program product define functions of the embodiments (including the methods described herein) and can be contained on a variety of computer-readable storage media. Illustrative computer-readable storage media include, but are not limited to: (i) non-writable storage media (e.g., read-only memory (ROM) devices within a computer, such as CD-ROM disks readably by a CD-ROM drive, flash memory, ROM chips, or any type of solid-state non-volatile memory) on which information is permanently stored; and (ii) writable storage media (e.g., floppy disks within a diskette drive or hard-disk drive or any type of solid state random-access memory) on which alterable information is stored. Such computer-readable storage media, when carrying computer-readable instructions that direct the functions of the disclosed embodiments, are embodiments of the present disclosure.

It will be appreciated to those skilled in the art that the preceding examples are exemplary and not limiting. It is intended that all permutations, enhancements, equivalents, and improvements thereto are apparent to those skilled in the art upon a reading of the specification and a study of the drawings are included within the true spirit and scope of the present disclosure. It is therefore intended that the following appended claims include all such modifications, permutations, and equivalents as fall within the true spirit and scope of these teachings.

What is claimed:

1. A method, comprising:

retrieving, by a computing system, historical event data for a plurality of sporting matches, wherein retrieving the historical event data comprises:

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receiving, by the computing system, broadcast video data corresponding to the plurality of sporting matches;
 deriving, by the computing system, player tracking data from the broadcast video data;
 augmenting, by the computing system, the player tracking data with the historical event data;
 generating, by the computing system, one or more attributes of a team and of at least one player on the team based on the historical event data;
 receiving, by the computing system, a request to simulate and view an alternative historical game outcome;
 simulating, by the computing system, the alternative historical game outcome based on the one or more attributes generated by the computing system; and
 outputting, by the computing system, a graphical representation of the simulation.

2. The method of claim 1, the method further comprising: generating, by the computing system, body pose data for the at least one player based on the historical event data.

3. The method of claim 2, wherein the simulating the alternative historical game outcome is based on the body pose data.

4. The method of claim 1, wherein the graphical representation includes at least one skeletal representation of the at least one player, and wherein the at least one skeletal representation is based on the one or more attributes.

5. The method of claim 1, wherein the one or more attributes include one or more player trajectories, one or more team styles, and one or more player roles.

6. The method of claim 1, wherein the request comprises substituting a player that was in a play of a historical game with a target player that was not in the play.

7. A non-transitory computer readable medium comprising one or more sequences of instructions, which, when executed by a processor, causes a computing system to perform operations comprising:

- retrieving historical event data for a plurality of sporting matches, wherein retrieving the historical event data comprises:
 - receiving broadcast video data corresponding to the plurality of sporting matches;
 - deriving player tracking data from the broadcast video data;
 - augmenting the player tracking data with the historical event data;
- generating one or more attributes of a team and of at least one player on the team based on the historical event data;
- receiving a request to simulate and view an alternative historical game outcome;
- simulating the alternative historical game outcome based on the one or more attributes generated by the computing system; and
- outputting a graphical representation of the simulation.

8. The non-transitory computer readable medium of claim 7, the operations further comprising:

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generating body pose data for the at least one player based on the historical event data.

9. The non-transitory computer readable medium of claim 8, wherein the simulating the alternative historical game outcome is based on the body pose data.

10. The non-transitory computer readable medium of claim 7, wherein the graphical representation includes at least one skeletal representation of the at least one player, and wherein the at least one skeletal representation is based on the one or more attributes.

11. The non-transitory computer readable medium of claim 7, wherein the one or more attributes include one or more player trajectories, one or more team styles, and one or more player roles.

12. The non-transitory computer readable medium of claim 7, wherein the request comprises substituting a player that was in a play of a historical game with a target player that was not in the play.

13. A system comprising:

- a processor; and
- a memory having programming instructions stored thereon, which, when executed by the processor, causes the system to perform operations comprising:
 - retrieving, by a computing system, historical event data for a plurality of sporting matches, wherein retrieving the historical event data comprises:
 - receiving, by the computing system, broadcast video data corresponding to the plurality of sporting matches;
 - deriving, by the computing system, player tracking data from the broadcast video data;
 - augmenting, by the computing system, the player tracking data with the historical event data
 - generating, by the computing system, one or more attributes of a team and of at least one player on the team based on the historical event data;
 - receiving, by the computing system, a request to simulate and view an alternative historical game outcome;
 - simulating, by the computing system, the alternative historical game outcome based on the one or more attributes generated by the computing system; and
 - outputting, by the computing system, a graphical representation of the simulation.

14. The system of claim 13, the operations further comprising:

- generating, by the computing system, body pose data for the at least one player based on the historical event data.

15. The system of claim 14, wherein the simulating the alternative historical game outcome is based on the body pose data.

16. The system of claim 13, wherein the graphical representation includes at least one skeletal representation of the at least one player, and wherein the at least one skeletal representation is based on the one or more attributes.

17. The system of claim 13, wherein the one or more attributes include one or more player trajectories, one or more team styles, and one or more player roles.

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